Lecture 9

Maxwell: variation and identity

An electromagnetic potential has four components, but its field equations cannot determine all four independently. We derive the equations from an action and identify the differential relation between them.

Objectives. *We fix the electromagnetic signs, vary the Maxwell action and derive charge conservation as a compatibility condition. We prove that gauge transformations leave the field unchanged and explain the resulting identity between equations.*

By the end of this lecture you will be able to:

- Derive Gauss's and Ampère's laws from the action with the stated signs.
- Prove the off-shell identity for the Maxwell equations.
- Identify the restriction on a prescribed current and the freedom in its potential.

9.1 Potential and electromagnetic field

We work on $\mathbb{R}^{1+3}$ with metric $\eta = \operatorname{diag}(1,-1,-1,-1)$ and orientation $\mathrm{vol} = \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$. All fields in this lecture are real.

Definition 9.1 (Gauge potential and gauge transformation)**.** An electromagnetic *gauge potential* is a C^2 one-form

$$A_{\mathrm{em}} = -\phi_{\mathrm{em}}\,\mathrm{d}t + A_x\,\mathrm{d}x + A_y\,\mathrm{d}y + A_z\,\mathrm{d}z.$$

Its *field strength* is the two-form $F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}$. A *gauge transformation*, with smooth scalar function χ, is

$$A_{\mathrm{em}} \longmapsto A_{\mathrm{em}} + \mathrm{d}\chi, \qquad \phi_{\mathrm{em}} \longmapsto \phi_{\mathrm{em}} - \chi_t, \qquad \mathbf{A} \longmapsto \mathbf{A} + \nabla\chi, \tag{9.1}$$

where $\mathbf{A} = (A_x, A_y, A_z)$ is the spatial potential.

Recall from *Topology and Geometry*, Part I, that the exterior derivative satisfies $\mathrm{d}^2 = 0$ and the graded product rule $\mathrm{d}(\alpha \wedge \beta) = \mathrm{d}\alpha \wedge \beta + (-1)^p \alpha \wedge \mathrm{d}\beta$ when α has degree p. Thus the transformed potential has the same field strength. This assertion uses equality of mixed derivatives, not a field equation.

Expanding $\mathrm{d}A_{\mathrm{em}}$ defines the electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ by

$$\begin{aligned} F_{\mathrm{em}} &= E_x\,\mathrm{d}x \wedge \mathrm{d}t + E_y\,\mathrm{d}y \wedge \mathrm{d}t + E_z\,\mathrm{d}z \wedge \mathrm{d}t \\ &\quad + B_x\,\mathrm{d}y \wedge \mathrm{d}z + B_y\,\mathrm{d}z \wedge \mathrm{d}x + B_z\,\mathrm{d}x \wedge \mathrm{d}y, \\ \mathbf{E} &= -\nabla\phi_{\mathrm{em}} - \partial_t \mathbf{A}, \qquad \mathbf{B} = \operatorname{curl}\mathbf{A}. \end{aligned} \tag{9.2}$$

The spatial dot product and curl use the Euclidean spatial metric. Since $\mathrm{d}F_{\mathrm{em}} = 0$, these fields obey $\operatorname{div}\mathbf{B} = 0$ and $\partial_t \mathbf{B} + \operatorname{curl}\mathbf{E} = 0$. For example, the coefficient of $\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$

is $\partial_x B_x + \partial_y B_y + \partial_z B_z$. The coefficient of $\mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z$ is $\partial_t B_x + \partial_y E_z - \partial_z E_y$. The other two coefficients follow by cyclic permutation.

9.2 The spacetime Hodge star

The action needs a map from two-forms to two-forms. We construct the needed constant-metric Hodge star here. This also recalls the convention of *Topology and Geometry*, Lecture 15. For basis wedges of degree p, the metric pairing is the determinant

$$\langle \mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_p}, \mathrm{d}x^{j_1} \wedge \cdots \wedge \mathrm{d}x^{j_p} \rangle = \det(\eta^{i_a j_b})_{a,b=1}^{p}.$$

Here $x^0 = t$, $x^1 = x$, $x^2 = y$ and $x^3 = z$.

Definition 9.2 (Spacetime Hodge star)**.** The *Hodge star* is the linear map from p-forms to $(4-p)$-forms determined by

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle \mathrm{vol} \qquad \text{for all forms } \alpha, \beta \text{ of degree } p. \tag{9.3}$$

An increasing basis wedge pairs under wedge product only with its complementary wedge. Its metric norm is the product of its diagonal metric signs. These two facts determine its star uniquely and construct the map. For example, $\mathrm{d}t \wedge \mathrm{d}x$ has metric norm squared -1, while $(\mathrm{d}t \wedge \mathrm{d}x) \wedge (\mathrm{d}y \wedge \mathrm{d}z) = \mathrm{vol}$. Hence $\star(\mathrm{d}t \wedge \mathrm{d}x) = -\mathrm{d}y \wedge \mathrm{d}z$. Applying the same rule gives all the entries needed below:

α	$\star\alpha$	β	$\star\beta$
$\mathrm{d}t \wedge \mathrm{d}x$	$-\mathrm{d}y \wedge \mathrm{d}z$	$\mathrm{d}y \wedge \mathrm{d}z$	$\mathrm{d}t \wedge \mathrm{d}x$
$\mathrm{d}t \wedge \mathrm{d}y$	$-\mathrm{d}z \wedge \mathrm{d}x$	$\mathrm{d}z \wedge \mathrm{d}x$	$\mathrm{d}t \wedge \mathrm{d}y$
$\mathrm{d}t \wedge \mathrm{d}z$	$-\mathrm{d}x \wedge \mathrm{d}y$	$\mathrm{d}x \wedge \mathrm{d}y$	$\mathrm{d}t \wedge \mathrm{d}z$

$$\star\mathrm{d}t = \mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z, \quad \star\mathrm{d}x = \mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z,$$
$$\star\mathrm{d}y = \mathrm{d}t \wedge \mathrm{d}z \wedge \mathrm{d}x, \quad \star\mathrm{d}z = \mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y.$$

The table gives $\star^2 = -1$ on two-forms. Symmetry of the determinant pairing also gives $\alpha \wedge \star\beta = \beta \wedge \star\alpha$ for equal degrees. In particular, varying $F_{\mathrm{em}} \wedge \star F_{\mathrm{em}}$ gives twice $\delta F_{\mathrm{em}} \wedge \star F_{\mathrm{em}}$.

9.3 Action and its variation

Let the prescribed charge density ρ and spatial current $\mathbf{J}$ be smooth functions of spacetime. Define the current one-form and current three-form by

$$\begin{aligned} j &= \rho\,\mathrm{d}t - J_x\,\mathrm{d}x - J_y\,\mathrm{d}y - J_z\,\mathrm{d}z, \qquad \mathcal{J} = \star j, \\ \mathcal{J} &= \rho\,\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z - J_x\,\mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z - J_y\,\mathrm{d}t \wedge \mathrm{d}z \wedge \mathrm{d}x - J_z\,\mathrm{d}t \wedge \mathrm{d}x \wedge \mathrm{d}y. \end{aligned} \tag{9.4}$$

Let $D \subset \mathbb{R}^{1+3}$ be a compact region, the closure of its interior, with the fields defined on an open neighbourhood of D. Only variations compactly supported in its interior will be used.

Definition 9.3 (Maxwell action)**.** The *Maxwell action* with prescribed current is

$$S_D[A_{\mathrm{em}}] = -\frac{1}{2}\int_D F_{\mathrm{em}} \wedge \star F_{\mathrm{em}} + \int_D A_{\mathrm{em}} \wedge \mathcal{J}, \qquad F_{\mathrm{em}} = \mathrm{d}A_{\mathrm{em}}. \tag{9.5}$$

The current is held fixed when the potential varies.

Proposition 9.4 (Maxwell variation and differential identity)**.** *For a C^2 potential and every smooth compactly supported one-form variation a in the interior of D,*

$$\delta S_D[A_{\mathrm{em}}]a = \int_D a \wedge (\mathcal{J} - \mathrm{d}{\star}F_{\mathrm{em}}). \tag{9.6}$$

Stationarity is equivalent to $\mathrm{d}{\star}F_{\mathrm{em}} = \mathcal{J}$. The off-shell identity $\mathrm{d}(\mathrm{d}{\star}F_{\mathrm{em}}) \equiv 0$ makes $\mathrm{d}\mathcal{J} = 0$ necessary for a solution. The action is invariant under every compactly supported gauge transformation exactly when $\mathrm{d}\mathcal{J} = 0$ in the interior of D. Gauge transformations leave the field equations unchanged and prevent them from determining the potential uniquely.

Proof. Since the metric is fixed, its star does not vary. The symmetry just proved and $\delta F_{\mathrm{em}} = \mathrm{d}a$ give

$$\delta S_D[A_{\mathrm{em}}]a = -\int_D \mathrm{d}a \wedge {\star}F_{\mathrm{em}} + \int_D a \wedge \mathcal{J}.$$

The graded product rule gives $\mathrm{d}(a \wedge {\star}F_{\mathrm{em}}) = \mathrm{d}a \wedge {\star}F_{\mathrm{em}} - a \wedge \mathrm{d}{\star}F_{\mathrm{em}}$. Thus the boundary contribution is the exact term $-\int_D \mathrm{d}(a \wedge {\star}F_{\mathrm{em}})$. It is zero: extending its compactly supported coefficients by zero, each coordinate derivative integrates to zero by Fubini and the fundamental theorem of calculus. The remaining terms give (9.6). The pairing of one-forms with three-forms by wedge product is nondegenerate. Testing each coefficient with the fundamental lemma of Lecture 1 proves the stationarity assertion. Its smooth-test version follows by choosing a smooth nonnegative bump in a ball where the coefficient keeps its sign.

Applying d to the field equation gives $\mathrm{d}\mathcal{J} = 0$ because $\mathrm{d}^2 = 0$. For a C^2 potential the coefficients of $\mathrm{d}{\star}F_{\mathrm{em}}$ are continuous. Its further derivative is interpreted distributionally, where mixed derivatives still commute. For a smooth potential the same identity holds pointwise. No equation for the potential was used to obtain the identity.

A gauge transformation leaves F_{em} unchanged. Its exact change in the action is therefore

$$S_D[A_{\mathrm{em}} + \mathrm{d}\chi] - S_D[A_{\mathrm{em}}] = \int_D \mathrm{d}\chi \wedge \mathcal{J} = -\int_D \chi\, \mathrm{d}\mathcal{J}, \qquad \chi \in C_c^\infty(\operatorname{int} D). \tag{9.7}$$

Here the discarded exact term is $\int_D \mathrm{d}(\chi\mathcal{J})$. The fundamental lemma makes invariance for every χ equivalent to $\mathrm{d}\mathcal{J} = 0$. Finally, adding any $\mathrm{d}\chi \neq 0$ preserves the field strength and hence every field equation, which proves the nonuniqueness. ∎

9.4 Gauss and Ampère from the scalar density

Example 9.5 (The component field equations)**.** Compute the Euler–Lagrange equations of the Maxwell action for ϕ_{em} and $\mathbf{A}$. The Hodge table and the complementary wedge signs give

$$F_{\mathrm{em}} \wedge {\star}F_{\mathrm{em}} = (|\mathbf{B}|^2 - |\mathbf{E}|^2)\mathrm{vol}, \qquad A_{\mathrm{em}} \wedge \mathcal{J} = (-\rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A})\mathrm{vol}.$$

Thus the scalar density is

$$L = \frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}|^2) - \rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A}.$$

The scalar potential occurs through $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \mathbf{A}_t$. Its density derivatives are $\partial L/\partial\phi_{\mathrm{em}} = -\rho$ and $\partial L/\partial(\partial_i\phi_{\mathrm{em}}) = -E_i$. The Euler–Lagrange expression is consequently

$$E_{\phi_{\mathrm{em}}} = \operatorname{div}\mathbf{E} - \rho.$$

For the spatial component A_x, the nonzero derivative coefficients are

$$\frac{\partial L}{\partial A_x} = J_x, \qquad \frac{\partial L}{\partial(\partial_t A_x)} = -E_x, \qquad \frac{\partial L}{\partial(\partial_y A_x)} = B_z, \qquad \frac{\partial L}{\partial(\partial_z A_x)} = -B_y.$$

They give $E_{A_x} = J_x + \partial_t E_x - \partial_y B_z + \partial_z B_y$. Cyclic permutation gives the other two components. Stationarity therefore produces Gauss's and Ampère's laws with the specified signs:

$$\operatorname{div}\mathbf{E} = \rho, \qquad \operatorname{curl}\mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}. \tag{9.8}$$

These equations agree with the form equation. Indeed, the spatial coefficient of $\mathrm{d}{\star}F_{\mathrm{em}}$ is $\operatorname{div}\mathbf{E}$. Its $\mathrm{d}t \wedge \mathrm{d}y \wedge \mathrm{d}z$ coefficient is $\partial_t E_x - (\operatorname{curl}\mathbf{B})_x$. Comparison with (9.4) gives the same source signs, and cyclic permutation supplies the rest.

9.5 The identity and the source restriction

The components of (9.4) give the conservation condition

$$\mathrm{d}\mathcal{J} = (\partial_t \rho + \operatorname{div}\mathbf{J})\mathrm{vol} = 0.$$

To see the identity directly in the equations, set $\mathcal{M}_0 = \operatorname{div}\mathbf{E}$ and $\boldsymbol{\mathcal{M}} = \partial_t \mathbf{E} - \operatorname{curl}\mathbf{B}$. Commutation of spatial derivatives gives $\operatorname{div}\operatorname{curl}\mathbf{B} = 0$, so

$$\partial_t \mathcal{M}_0 - \operatorname{div}\boldsymbol{\mathcal{M}} = \partial_t \operatorname{div}\mathbf{E} - \operatorname{div}(\partial_t \mathbf{E} - \operatorname{curl}\mathbf{B}) \equiv 0. \tag{9.9}$$

For C^2 potentials this equality is distributional. On a solution, $\mathcal{M}_0 = \rho$ and $\boldsymbol{\mathcal{M}} = -\mathbf{J}$, giving charge conservation.

This identity also follows from gauge invariance of the source-free action. For a smooth potential, its variation along $\delta\phi_{\mathrm{em}} = -\chi_t$ and $\delta\mathbf{A} = \nabla\chi$ is

$$0 = \int_D (-\mathcal{M}_0 \chi_t + \boldsymbol{\mathcal{M}} \cdot \nabla\chi)\,\mathrm{vol} = \int_D \chi(\partial_t \mathcal{M}_0 - \operatorname{div}\boldsymbol{\mathcal{M}})\,\mathrm{vol}.$$

Arbitrariness of the compactly supported function χ forces the identity. For C^2 potentials the same integration by parts defines its distributional meaning. This is the first instance of Noether's second theorem: a symmetry depending on an arbitrary function gives a differential relation between field equations. Lecture 19 develops the general statement.

Example 9.6 (An incompatible prescribed current). Take $\rho(t,x) = t$ and $\mathbf{J}(t,x) = 0$ on an open spacetime region. Its continuity expression is $\partial_t \rho + \operatorname{div}\mathbf{J} = 1$. If a potential solved Maxwell's equations there, the divergence of Ampère's law would give $\partial_t \operatorname{div}\mathbf{E} = 0$. The time derivative of Gauss's law would give $\partial_t \operatorname{div}\mathbf{E} = 1$. These contradict each other, even when interpreted distributionally. A nonconserved prescribed current is therefore inconsistent with these field equations. Lectures 10 and 13 require conserved sources.

Summary. The Maxwell action has density $\frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}|^2) - \rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A}$. Compactly supported variation gives $\mathrm{d}{\star}F_{\mathrm{em}} = \mathcal{J}$. The off-shell identity $\mathrm{d}^2{\star}F_{\mathrm{em}} = 0$ requires a conserved current. That condition also makes the source-coupled action gauge invariant for compactly supported gauge functions. Potentials related by $\mathrm{d}\chi$ give the same field. Lecture 10 fixes this freedom when solving the initial-data problem.

Self-check. Compute $\star(\mathrm{d}t \wedge \mathrm{d}x)$ and its star again. Which sign would an Euclidean metric incorrectly give in the first step?

Answer. The first result is $-\mathrm{d}y \wedge \mathrm{d}z$, and the second is $-\mathrm{d}t \wedge \mathrm{d}x$. An Euclidean metric would give $+\mathrm{d}y \wedge \mathrm{d}z$ in the first step.

Self-check. Start with the zero potential and use $\chi = tx^2 + yz$. Compute the transformed potential components and both fields.

Answer. The new components are $\phi_{\mathrm{em}} = -x^2$ and $\mathbf{A} = (2tx, z, y)$. The gradient and time derivative cancel in $\mathbf{E}$, so $\mathbf{E} = 0$. The curl is also zero, so $\mathbf{B} = 0$, although the potential is nonzero.

Self-check. For a gauge function supported in the interior of D, what is the exact change of the action if $\mathrm{d}\mathcal{J} \neq 0$?

Answer. It is $-\int_D \chi \, \mathrm{d}\mathcal{J}$. There is no boundary contribution because χ vanishes near the boundary. For nonzero $\mathrm{d}\mathcal{J}$, some such χ makes this change nonzero.

Self-check. Does charge conservation prove that Maxwell's equations determine the potential uniquely? Which identity holds before imposing any field equation?

Answer. No. Adding $\mathrm{d}\chi$ leaves F_{em} unchanged. The identity $\mathrm{d}(\mathrm{d}\star F_{\mathrm{em}}) = 0$ holds off shell, distributionally for a C^2 potential. Conservation is necessary for the equations to have a solution.